STUDY NOTES

CTET Paper II Science (English)

Complete 200 Practice MCQs (Content + Pedagogy)

2 MCQ/source file(s)
27 August 2026

200 MCQs

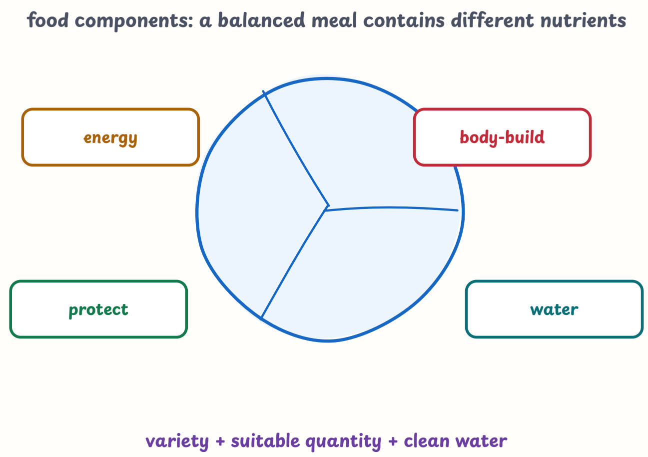
विषय-सूची / Contents

CTET Paper II Science MCQ

Part 1: Science Content — Q001-Q100

Set 01 — Food and Components of Food (Q001-Q010)

> Class VI-VIII science connects food components with functions, tests, health and diversity.



चित्र 1 — a balanced meal combines varied food components in suitable amounts

Q001. A balanced diet is one that—

- A. provides different nutrients, fibre and water in suitable amounts
- B. contains only one favourite food
- C. contains no fats or carbohydrates
- D. costs the most among all meals

Answer: A

Explanation: A balanced diet meets the body's varied needs; no single food supplies everything in the required proportion.

Q002. In the food figure, which component is mainly associated with growth and repair of body tissues?

- A. carbohydrate
- B. protein
- C. water only
- D. roughage only

Answer: B

Explanation: Proteins are body-building nutrients needed for growth and repair.

Q003. Deficiency of vitamin C may cause—

- A. rickets
- B. goitre
- C. scurvy
- D. night blindness

Answer: C

Explanation: Vitamin C deficiency is associated with scurvy; rickets is linked with vitamin D deficiency.

Q004. Iodine deficiency may lead to—

- A. anaemia
- B. scurvy
- C. beriberi
- D. goitre

Answer: D

Explanation: Iodine is needed for thyroid function; severe deficiency can cause enlargement of the thyroid gland called goitre.

Q005. The nutrient that is a major immediate source of energy is—

- A. carbohydrate
- B. protein only
- C. vitamin only
- D. mineral salt only

Answer: A

Explanation: Carbohydrates are commonly broken down to release energy for body activities.

Q006. Proteins are especially important for—

- A. making food taste sweet
- B. growth and repair
- C. forming only roughage
- D. reflecting light

Answer: B

Explanation: Proteins help build and repair body tissues.

Q007. When iodine solution turns a food sample blue-black, it indicates the presence of—

- A. protein
- B. fat only
- C. starch
- D. vitamin C

Answer: C

Explanation: Iodine gives a blue-black colour with starch.

Q008. Which is mainly an animal source of food?

- A. rice
- B. gram
- C. spinach
- D. milk

Answer: D

Explanation: Milk is obtained from animals; rice, gram and spinach are plant sources.

Q009. Roughage in food mainly helps to—

- A. support the movement of food through the intestine
- B. provide all body proteins
- C. replace clean water
- D. make bones luminous

Answer: A

Explanation: Dietary fibre adds bulk and supports healthy bowel movement, though it is not digested as a nutrient.

Q010. Which statement about healthy food habits is most appropriate?

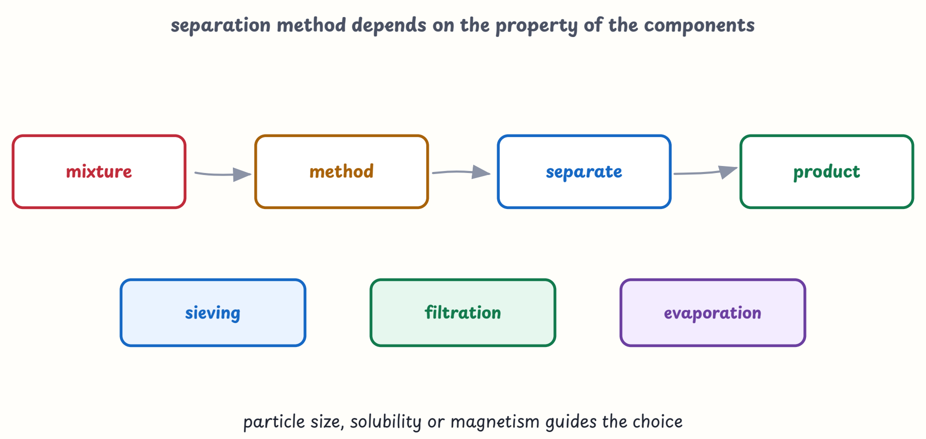
- A. Only expensive packaged food is healthy
- B. Food should be varied, clean, adequately cooked and taken in suitable quantities.
- C. Skipping water improves digestion
- D. The same food must be eaten by every person

Answer: B

Explanation: Health depends on variety, hygiene, preparation, quantity and individual needs.

Set 02 — Materials, Properties and Separation (Q011-Q020)

Materials are classified and separated by observable properties such as size, solubility, magnetism and state.



चित्र 2 — choose a separation method from the property of the mixture

Q011. The choice of a method for separating a mixture should mainly depend on—

- A. the colour of the container only
- B. the age of the person separating it
- C. the different properties of its components
- D. the number of letters in the material names

Answer: C

Explanation: Separation uses differences such as particle size, solubility, density or magnetic behaviour.

Q012. Handpicking is suitable for separating—

- A. salt dissolved in water
- B. iron filings from sand
- C. water from alcohol
- D. large visible stones from rice

Answer: D

Explanation: Handpicking works when unwanted components are relatively large and easily visible.

Q013. Filtration can separate—

- A. an insoluble solid from a liquid
- B. sugar from sugar solution without evaporation
- C. two completely mixed gases
- D. iron from a magnet

Answer: A

Explanation: A filter retains insoluble solid particles while the liquid passes through.

Q014. Which method is useful for obtaining salt from salt solution?

- A. handpicking
- B. evaporation
- C. magnetic separation
- D. sieving

Answer: B

Explanation: Evaporating water leaves the dissolved salt behind.

Q015. Iron filings can be separated from sand by using—

- A. a sieve only
- B. a filter paper only
- C. a magnet
- D. a thermometer

Answer: C

Explanation: Iron is attracted by a magnet while sand is not.

Q016. Which substance is soluble in water?

- A. sand
- B. sawdust
- C. small stones
- D. sugar

Answer: D

Explanation: Sugar dissolves in water; sand, sawdust and stones remain insoluble under ordinary conditions.

Q017. Burning a sheet of paper is generally a—

- A. chemical change
- B. reversible change
- C. change of position only
- D. change that forms paper again on cooling

Answer: A

Explanation: Burning forms new substances such as ash and gases and cannot normally be reversed to paper.

Q018. Melting an ice cube is a—

- A. chemical change forming a new substance
- B. reversible physical change
- C. permanent change in composition
- D. change caused only by magnetism

Answer: B

Explanation: Ice and water are the same substance in different physical states, so melting can be reversed by freezing.

Q019. Rusting of iron generally requires—

- A. sunlight only
- B. salt and darkness only
- C. oxygen and moisture
- D. carbon dioxide without water

Answer: C

Explanation: Iron rusts more readily when both oxygen and water are available.

Q020. Copper is commonly used for electric wires because it is—

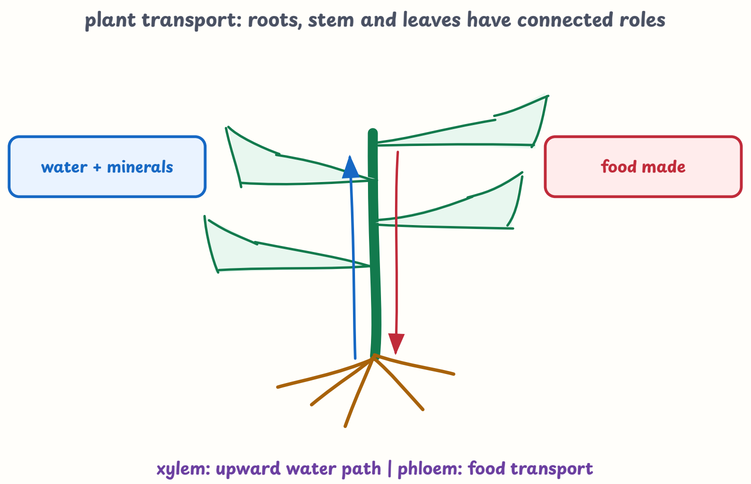
- A. transparent and magnetic
- B. a poor conductor of electricity
- C. soluble in air
- D. a good conductor and can be drawn into wires

Answer: D

Explanation: Copper conducts electricity well and is ductile, making it useful for wires.

Set 03 — Plants and Life Processes (Q021-Q030)

Plant structure is connected with transport, nutrition, reproduction and exchange with the environment.



चित्र 3 — water and minerals move upward while prepared food is transported to plant parts

Q021. In the plant figure, the upward movement of water and minerals from roots is mainly associated with—

- A. xylem
- B. phloem
- C. stomata only
- D. flower petals

Answer: A

Explanation: Xylem carries water and dissolved minerals upward from the roots; phloem transports prepared food.

Q022. The raw materials used by green plants for photosynthesis are mainly—

- A. oxygen and protein
- B. carbon dioxide and water
- C. nitrogen gas and starch
- D. only minerals and oxygen

Answer: B

Explanation: Plants use carbon dioxide and water in the presence of light and chlorophyll to make food.

Q023. Stomata mainly help a leaf in—

- A. absorbing food from soil
- B. anchoring the plant
- C. exchange of gases and loss of water vapour
- D. making seeds move

Answer: C

Explanation: Stomata are tiny openings involved in gas exchange and transpiration.

Q024. A seed generally needs water, air and a suitable temperature for—

- A. photosynthesis before it sprouts
- B. pollination by an insect
- C. formation of a fossil
- D. germination

Answer: D

Explanation: These conditions activate growth in a viable seed; light is not universally essential for the first stage of germination.

Q025. Potato can produce a new plant from its eyes. This is an example of—

- A. vegetative propagation
- B. pollination only
- C. seed dispersal by wind
- D. binary fission

Answer: A

Explanation: Potato buds or eyes can grow into new plants without the formation of a seed.

Q026. Pollination is the transfer of pollen from—

- A. root to stem
- B. anther to stigma
- C. leaf to root hair
- D. ovary to sepal only

Answer: B

Explanation: Pollen is produced in the anther and reaches the stigma during pollination.

Q027. Which plant commonly has a tap root?

- A. grass
- B. wheat
- C. carrot
- D. onion

Answer: C

Explanation: Carrot has a prominent main root; grasses and wheat typically have fibrous roots.

Q028. The green pigment that traps light energy in leaves is—

- A. haemoglobin
- B. iodine
- C. melanin
- D. chlorophyll

Answer: D

Explanation: Chlorophyll absorbs light energy needed for photosynthesis.

Q029. Transpiration is the loss of—

- A. water vapour from aerial parts of a plant
- B. oxygen from the roots only
- C. food from seeds
- D. soil particles from leaves

Answer: A

Explanation: Water vapour leaves mainly through stomata in the aerial parts of a plant.

Q030. Prepared food is transported from leaves to other parts mainly through—

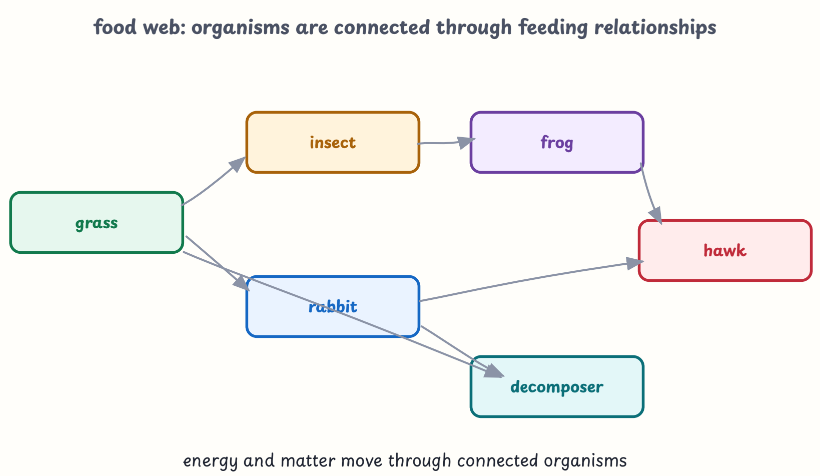
- A. xylem only
- B. phloem
- C. root cap
- D. stomata

Answer: B

Explanation: Phloem distributes food made in leaves to growing and storage regions.

Set 04 — Human Body, Animals and Adaptation (Q031–Q040)

Life processes can be studied through body systems, feeding relationships and adaptations.



चित्र 4 — several feeding links form a food web rather than one isolated chain

Q031. Digestion of food begins in the—

- A. stomach only
- B. small intestine only
- C. mouth
- D. large intestine

Answer: C

Explanation: Chewing and saliva begin the mechanical and chemical digestion of food in the mouth.

Q032. Most digested nutrients are absorbed into the blood in the—

- A. oesophagus
- B. large intestine only
- C. lungs
- D. small intestine

Answer: D

Explanation: The small intestine has structures that provide a large surface for absorption of digested nutrients.

Q033. Exchange of oxygen and carbon dioxide in humans occurs mainly in the—

- A. lungs
- B. stomach
- C. kidneys only
- D. bones

Answer: A

Explanation: The lungs contain surfaces where gases are exchanged between air and blood.

Q034. The pulse felt at the wrist is related to—

- A. movement of food in the intestine
- B. rhythmic pumping of the heart
- C. growth of hair
- D. evaporation of sweat only

Answer: B

Explanation: Each heartbeat sends a pressure wave through arteries that can be felt as a pulse.

Q035. The skeleton helps the body by—

- A. digesting all food
- B. making sunlight
- C. supporting the body, protecting organs and helping movement
- D. pumping air into leaves

Answer: C

Explanation: Bones provide a framework, protect organs and work with muscles for movement.

Q036. Vaccination mainly helps the body to—

- A. digest starch in the mouth
- B. increase height immediately
- C. replace the need for hygiene
- D. develop protection against a particular infection

Answer: D

Explanation: Vaccines train immune responses; they complement, rather than replace, hygiene and public-health measures.

Q037. In the food-web figure, an arrow from grass to a rabbit represents—

- A. transfer of food energy from producer to herbivore
- B. the rabbit making food for grass
- C. movement of water into the soil
- D. decomposition of a rock

Answer: A

Explanation: The rabbit obtains food energy by feeding on the plant, so the link goes from food to consumer.

Q038. Organisms such as fungi and many bacteria that break down dead matter are called—

- A. producers
- B. decomposers
- C. herbivores
- D. pollinators only

Answer: B

Explanation: Decomposers return simpler substances to the environment by breaking down dead material.

Q039. The hump of a camel mainly stores—

- A. extra water in a bottle-like sac
- B. air for breathing under sand
- C. fat
- D. undigested sand

Answer: C

Explanation: Camel humps store fat; adaptations such as concentrated urine help reduce water loss.

Q040. Scientists classify organisms mainly to—

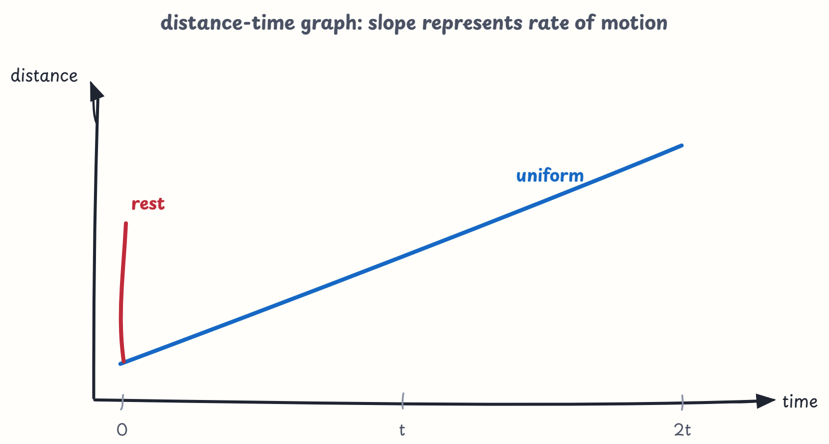
- A. prove that all organisms are identical
- B. remove the need for observation
- C. decide which organism is morally better
- D. organise them using shared characteristics and study relationships

Answer: D

Explanation: Classification makes the diversity of life easier to describe, compare and study.

Set 05 — Moving Things, Motion and Force (Q041–Q050)

Motion is described using distance, time and speed; force can change motion or shape.



चित्र 5 — a distance-time graph can show rest and uniform motion

Q041. Speed is calculated as—

- A. distance divided by time
- B. time divided by distance only
- C. distance multiplied by time
- D. distance plus time

Answer: A

Explanation: Speed tells how much distance is covered in a unit of time: speed = distance ÷ time.

Q042. A cyclist covers 120 km in 3 hours. What is the average speed?

- A. 30 km/h
- B. 40 km/h
- C. 360 km/h
- D. 123 km/h

Answer: B

Explanation: Average speed = 120 ÷ 3 = 40 km/h.

Q043. An object covers equal distances in equal intervals of time. Its motion is—

- A. randomly changing only
- B. impossible to describe
- C. uniform
- D. necessarily circular

Answer: C

Explanation: Equal distances in equal times describe uniform motion.

Q044. In the motion graph, the straight rising line represents—

- A. no passage of time
- B. an object moving backwards in time
- C. a permanent absence of motion
- D. distance increasing at a constant rate

Answer: D

Explanation: A straight distance-time line with constant slope represents uniform speed.

Q045. A horizontal line on a distance-time graph indicates that the object is—

- A. at rest during that interval
- B. moving with increasing speed
- C. covering infinite distance
- D. changing its mass

Answer: A

Explanation: If distance does not change as time passes, the object is at rest.

Q046. The motion of a pendulum repeating its to-and-fro path is—

- A. random motion only
- B. periodic motion
- C. rectilinear motion only
- D. a chemical change

Answer: B

Explanation: A motion that repeats itself after equal intervals is periodic.

Q047. A force can change an object's—

- A. chemical symbol only
- B. number of atoms in every case
- C. speed, direction or shape
- D. temperature never

Answer: C

Explanation: Pushes and pulls can alter motion and may deform an object.

Q048. Friction generally acts—

- A. only in the direction of motion
- B. only upward
- C. without contact in every situation
- D. opposite to the relative motion or attempted motion

Answer: D

Explanation: Friction resists sliding or the tendency to slide between surfaces.

Q049. Oil is applied between moving machine parts mainly to—

- A. reduce friction and wear
- B. increase roughness
- C. stop all forces
- D. make the parts magnetic

Answer: A

Explanation: A lubricant forms a separating layer and reduces frictional loss and damage.

Q050. Objects fall towards Earth because Earth exerts—

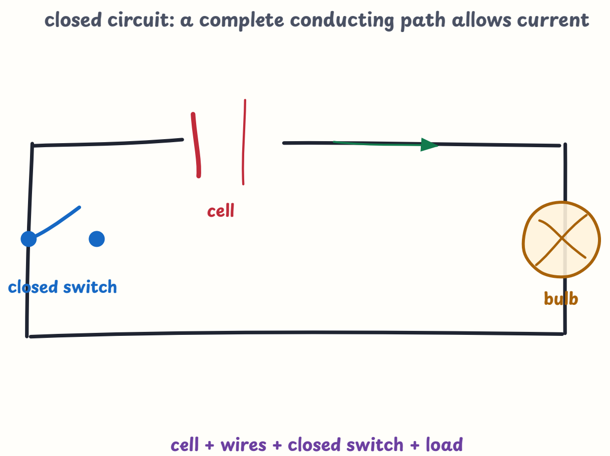
- A. magnetic force on every material
- B. gravitational force
- C. friction without contact
- D. sound force

Answer: B

Explanation: Gravity is the attractive force between Earth and objects near it.

Set 06 — Electric Current, Circuits and Magnets (Q051–Q060)

Electric circuits require a source, conductors and a complete path; magnets exert forces on particular materials.



चित्र 6 — a closed conducting path allows the bulb to receive current

Q051. In the circuit figure, the bulb can glow because there is—

- A. only a bulb without a cell
- B. a permanently broken wire
- C. a complete conducting path from the cell through the bulb
- D. an open switch in every branch

Answer: C

Explanation: Current can flow through the external circuit when the path is complete and conducting.

Q052. Which is a good conductor of electricity?

- A. rubber
- B. dry wood
- C. plastic
- D. copper

Answer: D

Explanation: Copper contains mobile charge carriers and is widely used in conducting wires.

Q053. Which material is generally an insulator?

- A. rubber
- B. copper
- C. aluminium
- D. iron

Answer: A

Explanation: Rubber resists the flow of electric current and is used to cover wires.

Q054. A cell in a simple circuit acts as a source of—

- A. only sound energy
- B. electrical energy
- C. soil nutrients
- D. water vapour

Answer: B

Explanation: A cell provides the energy that drives charge through the circuit.

Q055. When a switch is open in a simple circuit, the bulb generally—

- A. glows more brightly
- B. becomes a magnet automatically
- C. does not glow because the path is broken
- D. changes into a cell

Answer: C

Explanation: An open switch interrupts the conducting path.

Q056. A series circuit provides—

- A. many completely independent paths
- B. no path at all
- C. only a magnetic field without wires
- D. one continuous path for current

Answer: D

Explanation: In a simple series arrangement, components lie on one path.

Q057. When two north poles of magnets are brought near each other, they—

- A. repel each other
- B. attract strongly
- C. lose all magnetic properties
- D. turn into electric cells

Answer: A

Explanation: Like magnetic poles repel and unlike poles attract.

Q058. A freely suspended compass needle generally points approximately—

- A. east-west only
- B. north-south
- C. upward-downward only
- D. towards the nearest lamp

Answer: B

Explanation: Earth's magnetic field aligns the compass needle approximately along the north-south direction.

Q059. An electromagnet can be made stronger by—

- A. removing the current completely
- B. using only a plastic core with no coil
- C. increasing current or turns of wire around a suitable iron core
- D. breaking the conducting path

Answer: C

Explanation: A current-carrying coil, especially around soft iron, produces a temporary magnet whose strength can be varied.

Q060. A fuse protects a circuit because its wire—

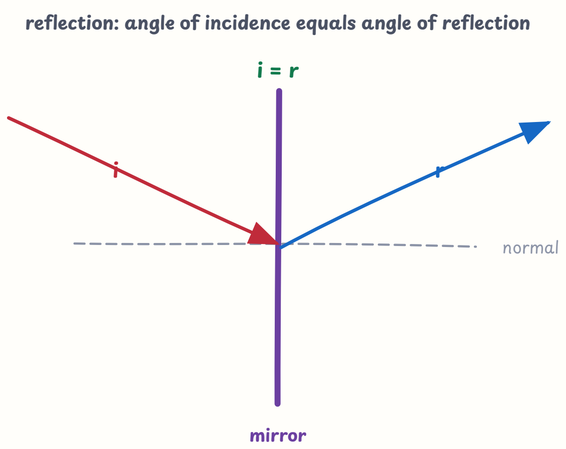
- A. always increases current
- B. stores drinking water
- C. reflects all light
- D. melts and breaks the circuit when excessive current flows

Answer: D

Explanation: A fuse is designed to melt during an overload and interrupt the circuit.

Set 07 — Light, Shadows and Natural Phenomena (Q061-Q070)

> Light travels in straight lines in a uniform medium and interacts with surfaces; natural phenomena need evidence-based safety explanations.



चित्र 7 — the incident and reflected rays make equal angles with the normal

Q061. According to the reflection figure, the angle of incidence is—

- A. equal to the angle of reflection
- B. always twice the angle of reflection
- C. always zero
- D. unrelated to the normal

Answer: A

Explanation: The law of reflection states that the two angles measured from the normal are equal.

Q062. Which is a luminous object?

- A. the Moon
- B. the Sun
- C. a book
- D. a mirror

Answer: B

Explanation: A luminous object emits its own light; the Moon, book and mirror are visible mainly through reflected light.

Q063. A dark shadow is most clearly formed when the object is—

- A. transparent
- B. a source of light
- C. opaque
- D. completely absent

Answer: C

Explanation: An opaque object blocks light and produces a distinct shadow.

Q064. A periscope works mainly by using—

- A. evaporation of water
- B. magnetic attraction only
- C. sound absorption
- D. reflection of light from mirrors

Answer: D

Explanation: Mirrors change the direction of light so an observer can see over an obstacle.

Q065. A prism can split white light into colours. This phenomenon is called—

- A. dispersion
- B. filtration
- C. condensation
- D. magnetisation

Answer: A

Explanation: Dispersion separates the component colours of white light.

Q066. Regular reflection is obtained most clearly from a—

- A. rough wall
- B. smooth, polished surface
- C. heap of sand
- D. cloud of smoke only

Answer: B

Explanation: A smooth surface reflects parallel incident rays in an orderly manner.

Q067. A solar or lunar eclipse requires the Sun, Earth and Moon to be—

- A. at random unrelated positions
- B. inside the same solid object
- C. approximately aligned
- D. all emitting their own light

Answer: C

Explanation: An eclipse occurs when one body comes into a particular alignment and casts a shadow.

Q068. The changing phases of the Moon are mainly due to—

- A. the Moon producing different amounts of light
- B. clouds permanently covering the Moon
- C. Earth changing the Moon's shape
- D. the changing portion of its sunlit half visible from Earth

Answer: D

Explanation: The Moon reflects sunlight; its position changes the visible fraction of the illuminated half.

Q069. During a thunderstorm, the safest choice is generally to—

- A. move indoors and avoid isolated trees and open fields
- B. stand under the tallest tree
- C. hold a metal pole in an open field
- D. lie beside a water tank outdoors

Answer: A

Explanation: A substantial building offers safer shelter than exposed locations or isolated tall objects.

Q070. Lightning is seen before thunder is heard because—

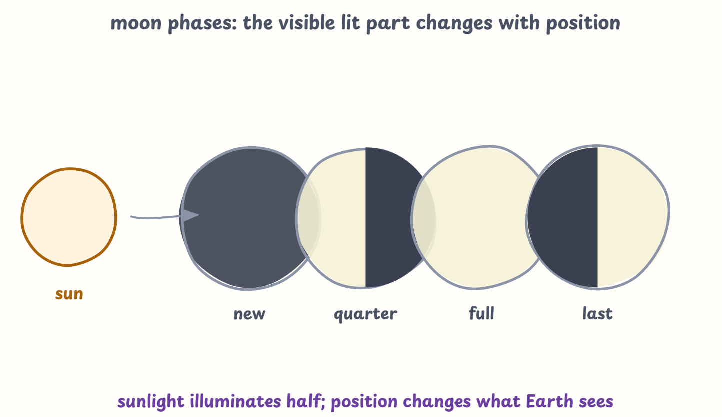
- A. sound cannot travel through air
- B. light travels much faster than sound
- C. thunder happens several minutes later
- D. clouds absorb all light

Answer: B

Explanation: Both signals are produced nearly together, but light reaches the observer sooner.

Set 08 — Weather, Water Cycle and the Sky (Q071-Q080)

Weather observations and the water cycle connect changes of state with atmospheric movement.



चित्र 8 — different positions show different visible portions of the Moon's lit half

Q071. In the Moon-phase figure, a full Moon is seen when—

- A. the Moon is between the Sun and Earth with its dark side away
- B. the Sun stops emitting light
- C. the Earth is approximately between the Sun and Moon
- D. the Moon produces its own light

Answer: C

Explanation: At full Moon, the side facing Earth is largely illuminated by sunlight; the simple alignment is Sun-Earth-Moon.

Q072. Evaporation from an open water surface is generally increased by—

- A. cooling and still air only
- B. covering the whole surface tightly
- C. removing all energy from the water
- D. heat, moving air and a larger exposed surface

Answer: D

Explanation: Higher temperature, wind and exposed surface give more water molecules opportunity to escape.

Q073. The change of water vapour into tiny liquid water droplets is—

- A. condensation
- B. evaporation
- C. melting
- D. freezing only

Answer: A

Explanation: Condensation is the change from gas to liquid, often seen in clouds or on a cool surface.

Q074. Which sequence is part of the water cycle?

- A. burning → rusting → melting
- B. evaporation → condensation → precipitation
- C. filtration → magnetism → digestion
- D. photosynthesis → pollination → eclipsing

Answer: B

Explanation: Water evaporates, vapour condenses and water may return as precipitation.

Q075. Humidity refers to the amount of—

- A. dust in soil only
- B. oxygen inside rocks
- C. water vapour present in the air
- D. salt dissolved in a river

Answer: C

Explanation: Humidity describes atmospheric water vapour.

Q076. Wind generally moves from a region of—

- A. lower pressure to higher pressure only
- B. zero pressure to solid rock
- C. equal pressure with no difference
- D. higher air pressure to lower air pressure

Answer: D

Explanation: Differences in air pressure drive the movement we experience as wind.

Q077. During the day, a sea breeze generally blows—

- A. from sea towards land
- B. from land towards sea only
- C. from the sky into the sea
- D. without any relation to heating

Answer: A

Explanation: Land heats faster than water, so warmer rising air over land is replaced by cooler air from the sea.

Q078. A rain gauge is used to measure—

- A. wind direction only
- B. amount of rainfall
- C. air pressure only
- D. soil temperature only

Answer: B

Explanation: A rain gauge collects and measures precipitation.

Q079. Weather forecasting becomes more useful when scientists—

- A. guess from one cloud only
- B. ignore instruments
- C. collect observations over time and analyse atmospheric data
- D. use no records

Answer: C

Explanation: Forecasts rely on measured patterns such as pressure, temperature, wind and humidity.

Q080. Which practice helps conserve water?

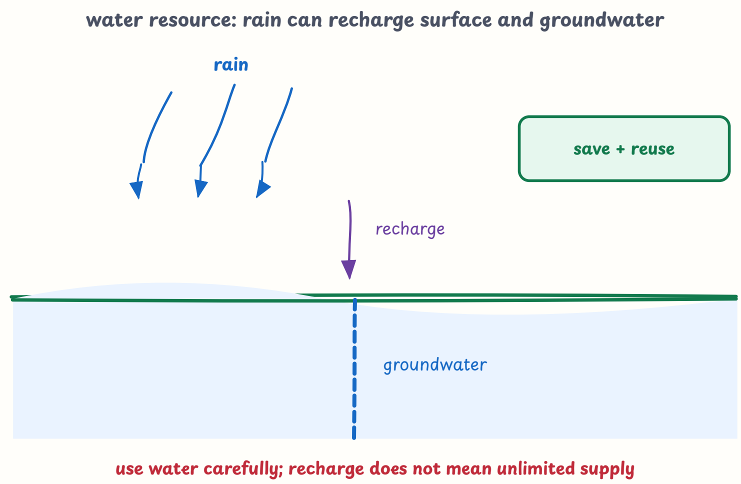
- A. leaving taps open
- B. washing roads with drinking water daily
- C. pumping groundwater without limit
- D. rainwater harvesting and repairing leaking taps

Answer: D

Explanation: Collecting rain and reducing leakage preserve available water resources.

Set 09 — Natural Resources and Environment (Q081–Q090)

Air, water, soil, organisms and energy resources are connected; sustainable use requires evidence and responsibility.



चित्र 9 — rainwater can recharge groundwater, but careful use is still necessary

Q081. The water-resource figure shows recharge. Groundwater recharge means—

- A. water enters and replenishes underground stores
- B. all groundwater becomes unlimited immediately
- C. water changes directly into coal
- D. rainwater is permanently removed from soil

Answer: A

Explanation: Infiltrating water can add to underground stores, although recharge is limited by soil, rocks and use.

Q082. Planting vegetation on a slope can reduce—

- A. photosynthesis
- B. soil erosion
- C. the formation of all clouds
- D. the need for sunlight

Answer: B

Explanation: Roots hold soil and vegetation slows the flow of water and wind across the surface.

Q083. Humus in soil generally—

- A. makes soil completely lifeless
- B. is a type of plastic
- C. improves fertility and helps the soil retain moisture
- D. prevents all air spaces

Answer: C

Explanation: Humus is decomposed organic matter that improves soil structure and nutrient availability.

Q084. Deforestation may lead to—

- A. more biodiversity in every case
- B. permanent increase in groundwater everywhere
- C. no change in local climate or soil
- D. soil erosion, habitat loss and changes in the water cycle

Answer: D

Explanation: Removing forests affects habitats, soil stability, water movement and local climate conditions.

Q085. Which is a renewable source of energy?

- A. solar energy
- B. coal
- C. petroleum
- D. natural gas

Answer: A

Explanation: Sunlight is replenished naturally on a human time scale; fossil fuels are finite.

Q086. Coal and petroleum are called non-renewable because—

- A. they are made every day by plants
- B. they form extremely slowly compared with their rate of use
- C. they cannot release energy
- D. they are unlimited in all regions

Answer: B

Explanation: Fossil fuels take geological time to form and can be depleted.

Q087. Treating sewage before releasing it helps to—

- A. increase untreated waste in rivers
- B. remove all need for clean water
- C. reduce pollution and protect water organisms and human health
- D. turn every pollutant into oxygen instantly

Answer: C

Explanation: Treatment removes or reduces harmful substances before wastewater enters the environment.

Q088. The three Rs of waste management are—

- A. read, repeat and rank
- B. remove, run and refill only
- C. react, rust and evaporate
- D. reduce, reuse and recycle

Answer: D

Explanation: The three Rs prioritise preventing waste, using items again and recovering materials.

Q089. In a food chain, arrows generally show—

- A. the direction of transfer of food energy
- B. the direction of rainfall only
- C. the age of each organism
- D. the size of an organism

Answer: A

Explanation: An arrow from food to consumer indicates transfer of energy through feeding.

Q090. Biodiversity means—

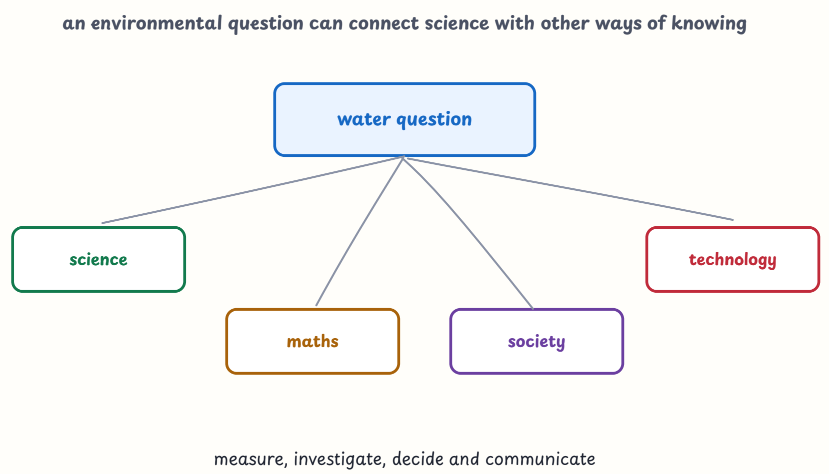
- A. only the number of trees in one park
- B. the variety of living organisms and ecosystems
- C. one species living everywhere
- D. the amount of sunlight in a room

Answer: B

Explanation: Biodiversity includes variation among organisms, species and ecosystems.

Set 10 — Science in Daily Life and Integrated Applications (Q091-Q100)

> Science concepts become useful when learners measure, explain, test and connect them with health, technology and society.



चित्र 10 — one environmental question can connect science, measurement, society and technology

Q091. The integrated-science figure best illustrates that an environmental question can be studied through—

- A. one isolated fact only
- B. guesswork without measurement
- C. connected scientific, mathematical, social and technological perspectives
- D. a subject with no real-life application

Answer: C

Explanation: Complex questions such as water use need evidence, measurement, social decisions and possible technologies.

Q092. A thermometer is used to measure—

- A. mass
- B. distance
- C. electric current only
- D. temperature

Answer: D

Explanation: Temperature is measured with a thermometer; different instruments measure different quantities.

Q093. Water fit for drinking without causing disease is called—

- A. potable water
- B. seawater in every case
- C. wastewater
- D. muddy water

Answer: A

Explanation: Potable water is sufficiently safe for drinking after appropriate treatment and testing.

Q094. Boiling drinking water mainly helps by—

- A. removing every dissolved salt
- B. killing many disease-causing microorganisms
- C. adding vitamins automatically
- D. turning water into a metal

Answer: B

Explanation: Adequate boiling can kill many microbes, though it does not remove all chemical contaminants.

Q095. A mosquito net helps prevent malaria mainly by—

- A. curing malaria after infection
- B. purifying drinking water
- C. reducing the chance of an infected mosquito biting a person
- D. increasing body temperature

Answer: C

Explanation: A net is a preventive barrier; it does not cure an infection.

Q096. Which waste is generally biodegradable?

- A. glass bottle
- B. aluminium foil
- C. plastic wrapper
- D. vegetable peels

Answer: D

Explanation: Vegetable peels can be decomposed by microorganisms, unlike many durable synthetic materials.

Q097. Burning plastic in the open is harmful mainly because it—

- A. can release pollutants and toxic fumes
- B. always improves air quality
- C. converts all waste to safe water
- D. adds oxygen without any residue

Answer: A

Explanation: Open burning can release harmful gases and particles and should be avoided.

Q098. A scientific explanation is stronger when it is based on—

- A. authority alone
- B. observations, evidence and a conclusion that can be checked
- C. a single unsupported opinion
- D. the most colourful diagram

Answer: B

Explanation: Evidence-based explanations are open to testing and revision.

Q099. Which action best reduces household energy use without changing the task?

- A. leaving appliances on standby forever
- B. opening the refrigerator repeatedly
- C. switching off unused lights and using efficient appliances
- D. burning extra fuel for the same work

Answer: C

Explanation: Reducing unnecessary use and improving efficiency conserve energy.

Q100. When a claim about a new water filter is made, a careful scientific response is to—

- A. accept the claim because it is printed on a poster
- B. test only once without recording conditions
- C. judge it by colour and advertisement
- D. test it with a fair procedure, record results and compare them with a suitable control

Answer: D

Explanation: A fair test with records and comparison provides stronger evidence about effectiveness.

Part 1 Quick Answer Index

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Revision target: Practise every concept with a “what do you observe?”, “why does it happen?” and “how would you test it?” question. Read units, arrows, circuit paths and graphs carefully.

CTET Paper II Science MCQ

Part 2: Science Pedagogy — Q101-Q200

Set 11 — Nature, Aims and Structure of Science (Q101–Q110)

Science teaching should build curiosity, evidence-based reasoning, explanation and appreciation of how knowledge is developed.

Q101. Science is best understood as—

- A. a way of asking questions, collecting evidence, explaining patterns and revising ideas
- B. a fixed list of facts to memorise
- C. only laboratory equipment
- D. a subject unrelated to everyday decisions

Answer: A

Explanation: Science includes knowledge as well as the processes and habits used to investigate the natural world.

Q102. A central aim of school science is to help learners—

- A. memorise names without relationships
- B. develop scientific literacy and use evidence to understand and respond to situations
- C. accept every claim from authority
- D. avoid observing the natural environment

Answer: B

Explanation: Scientific literacy supports understanding, reasoning, communication and responsible decision-making.

Q103. Scientific knowledge is generally—

- A. unchangeable after one observation
- B. based only on personal belief
- C. open to refinement when new evidence or better explanations become available
- D. true because a diagram is attractive

Answer: C

Explanation: Science is reliable because it uses evidence, but explanations can be revised as evidence improves.

Q104. A learner gives an incorrect explanation of evaporation. The teacher should first—

- A. punish the learner for speaking
- B. make the learner copy the definition only
- C. ignore the explanation and move to marks
- D. elicit the learner's idea, examine evidence and guide reconstruction

Answer: D

Explanation: Alternative conceptions are useful starting points for evidence-based conceptual change.

Q105. Why should science lessons connect with daily life?

- A. Familiar situations give concepts purpose and allow learners to apply evidence-based reasoning
- B. Daily life makes scientific accuracy unnecessary
- C. Only textbook examples can be scientific
- D. Context prevents learners from asking questions

Answer: A

Explanation: Applications help learners see why concepts matter while still requiring careful reasoning.

Q106. Which statement correctly distinguishes observation from inference?

- A. Observation and inference are identical
- B. Observation describes evidence noticed; inference is a reasoned interpretation of that evidence.
- C. Inference must always be a guess without evidence
- D. Observation can occur only in a laboratory

Answer: B

Explanation: Keeping the two separate helps learners record what they see before explaining it.

Q107. When a teacher encourages learners to ask their own "why" questions, the teacher mainly develops—

- A. only handwriting speed
- B. blind acceptance of facts
- C. curiosity and scientific inquiry
- D. fear of making observations

Answer: C

Explanation: Questioning is a starting point for investigation and explanation.

Q108. A safe science classroom requires the teacher to—

- A. ban all practical work
- B. allow learners to handle every chemical without instruction
- C. treat safety rules as optional
- D. anticipate hazards, explain procedures and supervise the responsible use of materials

Answer: D

Explanation: Safety enables practical learning; it is not the opposite of activity-based science.

Q109. The social nature of science is reflected when learners—

- A. share evidence, critique explanations and communicate findings with others
- B. work without recording any result
- C. accept only one person's private opinion
- D. avoid explaining how a conclusion was reached

Answer: A

Explanation: Science develops through communication, checking and collective reasoning as well as individual observation.

Q110. Which practice does NOT support scientific thinking?

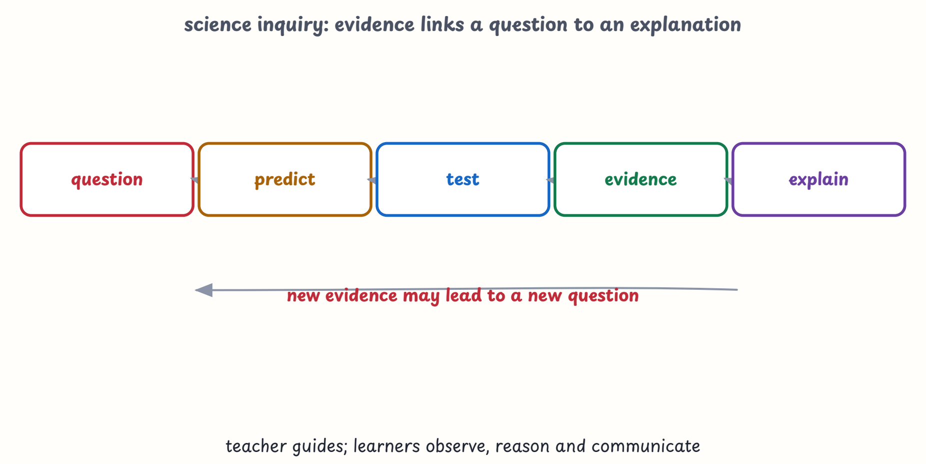
- A. comparing observations
- B. Telling learners to accept a conclusion without observing or examining evidence.
- C. repeating a fair test
- D. asking what evidence supports a claim

Answer: B

Explanation: Scientific thinking requires evidence and reasoning rather than unexamined acceptance.

Set 12 — Inquiry, Discovery and Classroom Approach (Q111-Q120)

Inquiry is not a free-for-all: the teacher structures a question, resources, safety, discussion and opportunities to revise explanations.



चित्र 1 — question, prediction, test, evidence and explanation form a revisable inquiry cycle

Q111. Which classroom task is most inquiry-oriented?

- A. Copy the definition of friction
- B. memorise a list of forces
- C. Investigate which surface produces the greatest friction, record evidence and explain the result.
- D. underline the word friction in a paragraph

Answer: C

Explanation: The task involves a question, investigation, evidence and explanation rather than recall alone.

Q112. A prediction in a science investigation should be—

- A. a conclusion written after seeing the result
- B. a random answer that cannot be tested
- C. a fact that never needs a condition
- D. a reasoned expectation that can be checked with evidence

Answer: D

Explanation: A prediction guides an investigation and can be supported, modified or rejected by evidence.

Q113. In an inquiry-based science lesson, the teacher's most appropriate role is to—

- A. design a safe learning situation, ask probing questions and guide evidence-based reasoning
- B. supply every conclusion before investigation
- C. remain absent while learners handle materials
- D. accept an answer without asking for evidence

Answer: A

Explanation: Teacher guidance and learner investigation work together in structured inquiry.

Q114. Which question is most suitable after learners observe that a wet cloth dries faster in sunlight?

- A. Who memorised the word evaporation?
- B. What variables might explain the difference, and how could we test them fairly?
- C. Is sunlight always the hottest object?
- D. Can the observation be ignored?

Answer: B

Explanation: The question moves from an observation to variables and a testable explanation.

Q115. A child predicts that a plant kept in darkness will grow exactly like one in light. The teacher should—

- A. announce the correct answer without investigation
- B. tell the child predictions are not allowed
- C. help the child design a comparison, identify conditions and observe the outcomes
- D. change the topic to avoid disagreement

Answer: C

Explanation: A test of the prediction turns a misconception or uncertainty into an opportunity for inquiry.

Q116. In a fair test of whether light affects plant growth, which factor should be changed deliberately?

- A. the type of plant, amount of water and light all together
- B. the measuring unit after every trial
- C. the conclusion before observing growth
- D. the amount or condition of light

Answer: D

Explanation: A fair test changes the independent variable while keeping relevant conditions controlled.

Q117. Why should a class repeat an investigation?

- A. Repeated trials help identify variation and make the evidence more dependable.
- B. One trial can never produce any information
- C. Repeating prevents learners from recording results
- D. It guarantees the preferred hypothesis is true

Answer: A

Explanation: Repetition does not guarantee a desired result; it improves confidence and exposes variation.

Q118. During a science discussion, a learner disagrees with a conclusion. The teacher should ask the learner to—

- A. remain silent because disagreement is disruptive
- B. state the evidence and reasoning supporting the alternative explanation
- C. choose the longest answer
- D. replace evidence with a personal insult

Answer: B

Explanation: Scientific disagreement is productive when it is respectful and evidence-based.

Q119. A well-planned field visit should include—

- A. only travel and entertainment
- B. no recording because memory is sufficient
- C. a question or purpose, observation tasks, safety planning and post-visit analysis
- D. a fixed conclusion given before the visit

Answer: C

Explanation: Fieldwork becomes learning when experience is organised around observation and reflection.

Q120. The inquiry-cycle figure is best summarised as—

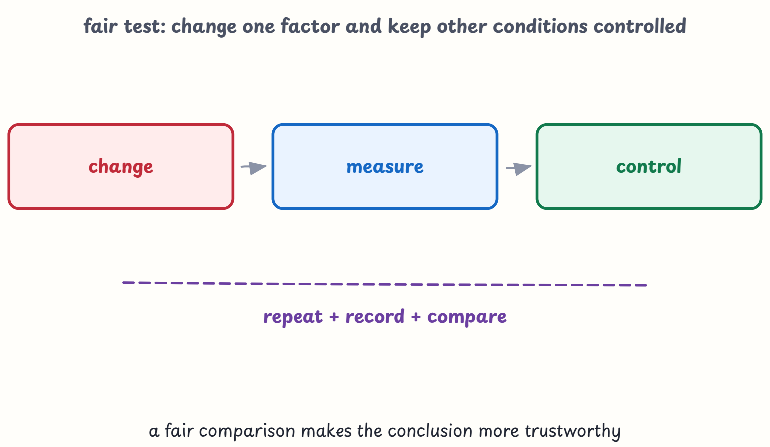
- A. memorise → copy → rank → stop
- B. observe nothing → guess → announce
- C. read a definition → avoid testing → forget
- D. question → predict → test → collect evidence → explain and revise

Answer: D

Explanation: Inquiry links questions and explanations through evidence and allows ideas to be revised.

Set 13 — Observation, Experimentation and Evidence (Q121-Q130)

> Learners need practice in careful observation, measurement, fair comparison, recording and interpreting evidence.



चित्र 2 — a fair test changes one factor, measures an outcome and controls other conditions

Q121. Which is an observation rather than an inference?

- A. The liquid has changed from clear to cloudy.
- B. A chemical reaction must have occurred
- C. The liquid is dangerous because it looks interesting
- D. The particles are definitely disappearing

Answer: A

Explanation: The first statement records what is directly noticed; the others interpret or speculate about it.

Q122. A fair test of which paper towel absorbs more water should change—

- A. paper type, water volume and time all at once
- B. the brand or type of paper towel while keeping water volume and procedure comparable
- C. the result after seeing it
- D. only the names written in the table

Answer: B

Explanation: Changing one relevant factor makes a comparison interpretable.

Q123. A variable kept the same in an investigation is called a—

- A. conclusion
- B. random error
- C. controlled variable
- D. dependent claim

Answer: C

Explanation: Controlled variables are held constant so that the effect of the chosen factor can be examined.

Q124. Why are standard units and suitable instruments important in a science investigation?

- A. They remove the need to observe
- B. They guarantee every hypothesis is correct
- C. They make recording results unnecessary
- D. They make measurements clearer, comparable and less dependent on personal estimates.

Answer: D

Explanation: Common units and instruments improve communication and comparison of evidence.

Q125. If a thermometer is read at an angle, the reading may be affected by—

- A. parallax error
- B. photosynthesis
- C. magnetic attraction
- D. condensation only

Answer: A

Explanation: Viewing a scale from the wrong position can make the apparent reading differ from the actual level.

Q126. Which record is most useful for comparing plant growth over five days?

- A. a memory-based final statement
- B. a dated table with the same measurement method used each day
- C. unlabelled drawings of different sizes
- D. only the tallest plant's name

Answer: B

Explanation: Dated, consistent records allow change and comparisons to be examined.

Q127. A conclusion in science should—

- A. repeat the prediction regardless of results
- B. include only an unsupported opinion
- C. answer the question using the recorded evidence and mention relevant limitations
- D. be decided before the test

Answer: C

Explanation: A conclusion interprets evidence and should not hide conditions or limitations.

Q128. Before a practical activity with glassware and a heat source, the teacher should—

- A. allow learners to discover every hazard by accident
- B. remove all instructions
- C. ask learners to work near loose wires and water
- D. demonstrate safe handling, explain hazards and organise supervision

Answer: D

Explanation: Safety planning is part of practical scientific competence.

Q129. A scientific model is useful when it—

- A. represents important relationships while its limits are made clear
- B. copies every detail of reality
- C. is accepted without comparison to evidence
- D. is judged only by decoration

Answer: A

Explanation: Models simplify reality; their usefulness depends on what they represent and where they do not apply.

Q130. The variables figure reminds a teacher to—

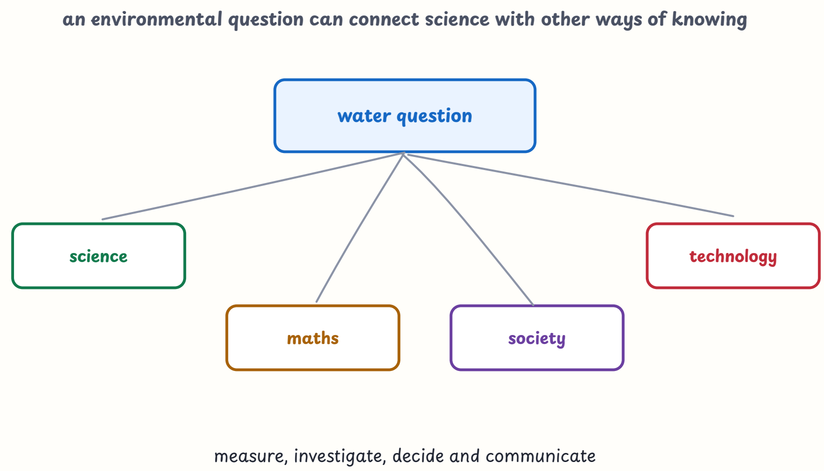
- A. change every condition at once
- B. change one factor, measure an outcome, control other conditions and repeat where possible
- C. choose a conclusion before testing
- D. avoid recording the procedure

Answer: B

Explanation: This structure supports a fair test and a more trustworthy conclusion.

Set 14 — Integrated and Contextual Science (Q131–Q140)

> Science learning can connect with mathematics, society, technology, language and local environmental concerns.



चित्र 3 — an environmental question can be investigated through connected perspectives

Q131. An integrated approach to science teaching means—

- A. mixing unrelated activities without a goal
- B. removing scientific accuracy
- C. connecting a scientific concept with related evidence, mathematics, society, technology or language
- D. teaching only one subject name at a time

Answer: C

Explanation: Integration is purposeful when different perspectives deepen understanding of a meaningful question.

Q132. A school water-use project can appropriately connect science with mathematics by asking learners to—

- A. draw water pictures without quantities
- B. avoid collecting any observations
- C. replace evidence with a slogan only
- D. measure use, organise data and estimate how a conservation change affects consumption

Answer: D

Explanation: Measurement and data help learners quantify a scientific and environmental issue.

Q133. Science and society are connected because scientific decisions—

- A. can affect health, resources, technology, work and the environment
- B. are made without values or consequences
- C. belong only inside laboratories
- D. never require communication with people

Answer: A

Explanation: Science is applied in social contexts and its uses have consequences that citizens consider.

Q134. A digital simulation is most useful in science class when it—

- A. replaces every hands-on experience automatically
- B. makes an inaccessible process visible and is followed by discussion or evidence-based tasks
- C. is shown only as entertainment
- D. is accepted as reality without checking assumptions

Answer: B

Explanation: Simulations are models; purposeful use includes interpretation and awareness of their limits.

Q135. Using a local pond to study water quality is valuable because it—

- A. means local water needs no testing
- B. makes textbook concepts unnecessary
- C. links scientific ideas with learners' environment and invites authentic observation
- D. guarantees that all ponds have identical conditions

Answer: C

Explanation: Local contexts motivate investigation while still requiring careful sampling and evidence.

Q136. Which activity connects science, language and art meaningfully?

- A. decorate a page without observing leaves
- B. copy a poem and call it an experiment
- C. avoid discussing what the labels represent
- D. Observe leaf patterns, record descriptions, read a short text and create a labelled representation with evidence.

Answer: D

Explanation: Different modes communicate evidence about the same scientific observation.

Q137. A lesson on air pollution should encourage learners to—

- A. identify sources, examine evidence, discuss effects and consider feasible responses
- B. memorise the word pollution only
- C. blame one person without data
- D. assume every visible cloud is pollution

Answer: A

Explanation: Environmental learning combines scientific understanding with evidence-based and responsible action.

Q138. The science-technology-society perspective asks learners to consider—

- A. only the design of a machine
- B. how scientific knowledge and technologies interact with human needs, choices and consequences
- C. science as completely separate from people
- D. a technology's advertisement as proof

Answer: B

Explanation: The perspective develops informed judgement about applications and their effects.

Q139. Which classroom question best promotes critical thinking about a technology?

- A. What is the product's colour?
- B. Who can memorise its slogan?
- C. What problem does it solve, what evidence supports it and what limitations or impacts might it have?
- D. Why should every technology be accepted?

Answer: C

Explanation: Critical thinking examines purpose, evidence, limitations and consequences.

Q140. The integrated-science figure can best be used to—

- A. teach four isolated definitions
- B. avoid subject-specific reasoning
- C. make learners copy the diagram without interpreting it
- D. start a shared question and assign measurement, investigation, social discussion and communication tasks

Answer: D

Explanation: An integrated figure is a prompt for connected inquiry, not a substitute for evidence.

Set 15 — Textbooks, Teaching Aids and Safe Resources (Q141-Q150)

> Materials should be selected for the concept, learner access, safety and opportunities for active reasoning.

Q141. A low-cost science teaching aid is most valuable when it—

- A. helps learners observe or model a concept and discuss the evidence
- B. looks expensive regardless of use
- C. is displayed but never handled or interpreted
- D. replaces the learning objective

Answer: A

Explanation: A simple material can be powerful when it makes a relationship visible and supports reasoning.

Q142. The role of a science textbook should be to—

- A. serve as the only source of experience
- B. provide organised concepts, questions, representations and references that teachers and learners examine critically
- C. replace observation and discussion
- D. be memorised without checking examples

Answer: B

Explanation: Textbooks are important resources but should work with inquiry, practical work and other materials.

Q143. A teacher demonstration is different from a learner experiment because—

- A. demonstrations always produce better learning
- B. experiments need no safety planning
- C. in a demonstration the teacher controls the action, while an experiment gives learners a planned opportunity to investigate and record
- D. learners cannot reason from demonstrations

Answer: C

Explanation: Both can be useful, but participation, control and evidence collection differ.

Q144. A simulation of the solar system should be accompanied by—

- A. the belief that it is the actual solar system
- B. no opportunity to interpret it
- C. a ban on observation of the sky
- D. questions about scale, assumptions and what the model can or cannot represent

Answer: D

Explanation: Models simplify reality, so learners should examine their scope and limitations.

Q145. A concept map in science helps learners—

- A. organise relationships among concepts and identify gaps or misconceptions
- B. memorise isolated terms without links
- C. avoid explaining connections
- D. measure only handwriting speed

Answer: A

Explanation: Concept maps make conceptual structure and relationships visible.

Q146. A model of the respiratory system is most useful when learners—

- A. treat every part as identical to a human organ
- B. use it to explain a process, identify what it represents and discuss its limitations
- C. decorate it without asking questions
- D. copy its labels without relating them to breathing

Answer: B

Explanation: Models support explanation when their representation and limits are made explicit.

Q147. An accessible science worksheet should—

- A. use tiny print and dense unexplained vocabulary
- B. assess only copying
- C. use readable layout, clear language, appropriate visuals and more than one possible way to show understanding when suitable
- D. exclude learners who need tactile or oral support

Answer: C

Explanation: Accessibility removes barriers while preserving the intended scientific goal.

Q148. Which practice avoids overdependence on polished laboratory equipment?

- A. cancel every practical activity without a laboratory
- B. pretend local materials have no limitations
- C. let equipment perform reasoning for learners
- D. Use safe local materials, learner observation, careful recording and discussion of limitations.

Answer: D

Explanation: Scientific inquiry can be developed with simple resources when procedures and evidence are sound.

Q149. When using a flame, chemical or sharp instrument in class, the teacher must—

- A. plan risk controls, explain safe handling and supervise the activity
- B. assume learners know every hazard
- C. allow unsafe mixing for excitement
- D. ignore protective equipment

Answer: A

Explanation: Safety is an essential condition of meaningful practical science.

Q150. The best basis for selecting a science teaching aid is—

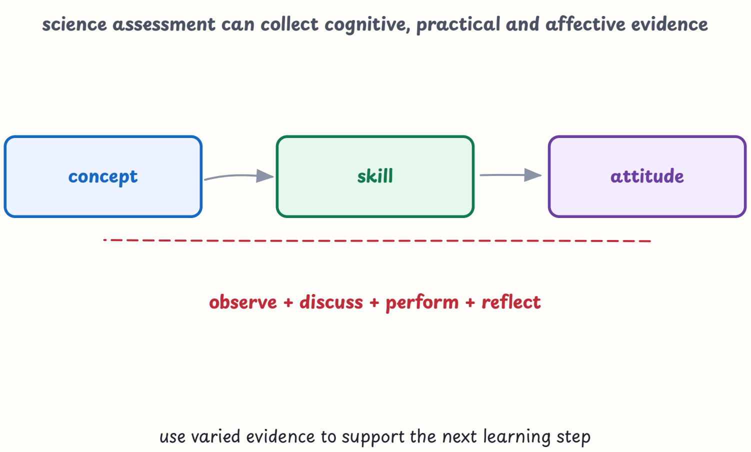
- A. its novelty alone
- B. its alignment with the learning goal, learner access, safety and potential for active use
- C. its price alone
- D. whether it makes the teacher's talk longer

Answer: B

Explanation: A resource should serve learning, not become an end in itself.

Set 16 — Evaluation of Science Learning (Q151–Q160)

Science assessment should include conceptual, practical and affective dimensions and should guide the next learning step.



चित्र 4 — science assessment can collect conceptual, practical and attitude evidence

Q151. Which task mainly assesses the cognitive dimension of science learning?

- A. Handle a measuring cylinder safely
- B. show care for a class plant
- C. Explain why a shadow changes length during the day using a labelled diagram.
- D. arrange laboratory stools

Answer: C

Explanation: The task asks for conceptual explanation and representation.

Q152. Which evidence mainly assesses a psychomotor science skill?

- A. state the definition of current
- B. express interest in wildlife
- C. write the date on a notebook
- D. Set up a simple circuit safely and make a correct observation of the bulb.

Answer: D

Explanation: Psychomotor assessment concerns coordinated practical performance.

Q153. Which evidence mainly assesses the affective dimension?

- A. Consistently handle organisms respectfully and show willingness to conserve resources.
- B. calculate speed from distance and time
- C. label the parts of a cell
- D. measure a liquid with a ruler

Answer: A

Explanation: Attitudes, values and responsible dispositions belong to the affective domain.

Q154. A teacher asks learners to draw a prediction before an investigation and uses responses to plan the next discussion. This is—

- A. only final certification
- B. formative assessment
- C. an intelligence label
- D. assessment with no teaching purpose

Answer: B

Explanation: Evidence gathered during learning is used to shape the next teaching step.

Q155. A rubric for a science investigation should include criteria such as—

- A. only the neatness of the cover
- B. only the final answer with no process
- C. question, procedure, observation, evidence-based conclusion and communication
- D. the learner's family occupation

Answer: C

Explanation: A relevant rubric assesses the process and quality of scientific work.

Q156. A science portfolio may contain—

- A. only the highest test mark
- B. only copied textbook pages
- C. only a list of absent days
- D. dated observations, drawings, data tables, drafts, reflections and revised explanations

Answer: D

Explanation: A portfolio documents development across different forms of scientific work.

Q157. An observation checklist during practical work should record—

- A. specific behaviours such as safe handling, measuring, recording and explaining
- B. a vague label such as clever
- C. only who finished first
- D. the learner's popularity

Answer: A

Explanation: Specific observable criteria produce useful evidence for feedback.

Q158. Which is an authentic science assessment task?

- A. copy the word conservation fifty times
- B. Use local water data to explain a problem and propose a justified conservation action.
- C. recite a definition without context
- D. circle every third noun in a paragraph

Answer: B

Explanation: Authentic tasks resemble purposeful scientific reasoning and communication.

Q159. A science test has validity when it—

- A. uses the most difficult vocabulary possible
- B. takes the longest time
- C. provides evidence about the intended scientific understanding or skill rather than irrelevant barriers
- D. gives identical results to every learner

Answer: C

Explanation: Validity concerns whether the assessment measures the construct it claims to measure.

Q160. The assessment-science figure reminds the teacher to—

- A. use only one written test
- B. ignore practical performance
- C. grade interest from a single smile
- D. collect evidence of concepts, practical skills and attitudes through varied tasks

Answer: D

Explanation: Science learning has multiple dimensions and needs appropriate evidence for each.

Set 17 — Misconceptions and Problems in Science Learning (Q161–Q170)

Children's alternative ideas are starting points for inquiry; effective teaching makes them visible and tests them with evidence.

Q161. An alternative conception is—

- A. a learner's existing explanation that may differ from accepted scientific understanding
- B. a deliberate act of disobedience
- C. proof of low intelligence
- D. a scientific law that needs no evidence

Answer: A

Explanation: Learners interpret experiences using prior ideas; teaching can build, refine or reconstruct them.

Q162. A learner says a heavy stone falls faster than a light stone in every situation. The teacher should—

- A. mock the learner's idea
- B. ask for the reasoning and design a safe comparison that examines the claim
- C. give only the correct sentence to copy
- D. avoid all discussion of falling objects

Answer: B

Explanation: Evidence and comparison can challenge an overgeneralisation more productively than ridicule.

Q163. A learner believes plants get all their food directly from soil. The best next step is to —

- A. say soil is never important
- B. ask the learner to memorise photosynthesis immediately
- C. use a plant-growth investigation and evidence about air, light, water and soil roles
- D. avoid discussing plant nutrition

Answer: C

Explanation: Investigation can differentiate resources needed for growth from the food produced by green plants.

Q164. A learner explains seasons only by saying Earth is much closer to the Sun in summer. The teacher should—

- A. accept the explanation because it sounds logical
- B. punish the learner for using a model
- C. teach a slogan without comparison
- D. use a model and evidence to discuss Earth's tilt, sunlight angle and seasonal patterns

Answer: D

Explanation: Models and evidence help examine a plausible but incomplete explanation.

Q165. A learner thinks a bulb glows with one wire connected to a cell. The teacher should—

- A. use circuit materials to test open and closed paths and ask the learner to trace the route
- B. give the circuit definition only
- C. say electricity is invisible so it cannot be studied
- D. allow unsafe short circuits

Answer: A

Explanation: A working model makes the need for a complete conducting path observable.

Q166. Why should a teacher elicit learners' ideas before teaching a science concept?

- A. It allows the teacher to label learners permanently
- B. It reveals prior reasoning and helps the teacher choose experiences that address specific needs.
- C. It replaces the need for evidence
- D. It ensures every learner has the same idea

Answer: B

Explanation: Prior ideas influence interpretation, so making them visible supports responsive teaching.

Q167. Cognitive conflict is productive when—

- A. the teacher deliberately humiliates a learner
- B. confusion is left without discussion
- C. evidence challenges an existing explanation and learners have support to reconstruct it
- D. a wrong answer is punished immediately

Answer: C

Explanation: A supported mismatch between idea and evidence can motivate conceptual change.

Q168. If learners misunderstand a science question because of difficult language, the teacher should—

- A. assume they lack scientific ability
- B. remove all reading from science
- C. give the answer before they reason
- D. clarify vocabulary and use diagrams or home-language support while preserving the science goal

Answer: D

Explanation: Language scaffolding separates an access barrier from the intended scientific concept.

Q169. Why is feedback saying only "wrong" inadequate in science?

- A. It does not identify the reasoning or evidence that needs revision.
- B. It is too specific
- C. It automatically develops inquiry
- D. It assesses practical skills completely

Answer: A

Explanation: Effective feedback points to evidence, reasoning and a next step.

Q170. When a misconception appears in many learners' answers, the teacher should—

- A. blame all learners for carelessness
- B. reconsider the representation or experience and design a shared conceptual intervention
- C. skip the topic permanently
- D. repeat the same explanation unchanged

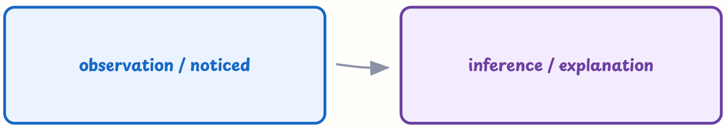
Answer: B

Explanation: A common pattern can signal a teaching or representation issue as well as a learner difficulty.

Set 18 — Inclusive, Differentiated and Remedial Science (Q171–Q180)

Every learner should have meaningful access to scientific observation, reasoning, practical work and communication.

observation and inference are related but not identical



record evidence first; explain with reasons and alternatives

a conclusion should be open to checking

चित्र 5 — record what is noticed before interpreting it with evidence

Q171. The first step in remedial science teaching should be—

- A. give the same worksheet to everyone
- B. lower the learning goal immediately
- C. diagnose the specific conceptual, language, practical or participation barrier
- D. label the learner weak

Answer: C

Explanation: Targeted support depends on knowing what is preventing access or understanding.

Q172. A learner cannot distinguish observation from inference. Which activity is most suitable?

- A. copy both definitions repeatedly
- B. avoid all experiments
- C. ask the learner to memorise the teacher's examples
- D. Sort statements into noticed evidence and possible explanation, then justify the sorting.

Answer: D

Explanation: Sorting and explaining makes the distinction concrete and supports scientific recording.

Q173. Differentiation in a science investigation may involve—

- A. different scaffolds or response modes while learners work toward a meaningful common concept
- B. giving unrelated science to some learners
- C. excluding learners from practical work
- D. lowering every expectation to copying

Answer: A

Explanation: Differentiation responds to learner variability without removing worthwhile science.

Q174. For a learner with visual impairment studying plant parts, an inclusive teacher can provide —

- A. only a small printed diagram
- B. tactile models, real specimens where safe, verbal descriptions and accessible recording
- C. exemption from all plant learning
- D. a test based only on visual copying

Answer: B

Explanation: Tactile and verbal resources make structure and function accessible.

Q175. An advanced learner in a science class should be offered—

- A. only more pages of copying
- B. permanent exemption from group work
- C. an open investigation involving variable control, evidence, explanation and reflection
- D. the role of teacher for every activity

Answer: C

Explanation: Enrichment deepens inquiry and reasoning rather than merely increasing repetition.

Q176. A multilingual learner uses a home-language term while explaining a scientific idea. The teacher should—

- A. reject the explanation before understanding it
- B. ban the home language
- C. assume the learner has no science concept
- D. value the idea, connect the term with the school-language word and continue the evidence-based discussion

Answer: D

Explanation: Language bridging preserves conceptual participation while developing scientific vocabulary.

Q177. If boys routinely handle apparatus while girls only record, the teacher should—

- A. rotate roles and ensure every learner practises both practical and communication skills
- B. accept the pattern as natural
- C. give boys extra marks for handling
- D. remove apparatus work from the lesson

Answer: A

Explanation: Equitable role rotation challenges stereotypes and distributes scientific opportunity.

Q178. A learner is anxious about practical science. The most appropriate support is to—

- A. force an unprepared public performance
- B. explain the procedure, begin with a low-risk role, allow rehearsal and build participation gradually
- C. remove the learner from all activities
- D. treat anxiety as misbehaviour

Answer: B

Explanation: Predictability, safety and gradual success support participation and confidence.

Q179. Family and community collaboration can strengthen science learning by—

- A. transferring all teaching responsibility to families
- B. requiring one language at home
- C. connecting school investigations with local knowledge, resources and environmental questions
- D. replacing classroom evidence with opinion

Answer: C

Explanation: Community knowledge is a resource when it is examined respectfully and connected with evidence.

Q180. The observation-inference figure can support inclusion by asking learners to—

- A. accept a single interpretation without observation
- B. assess only fluent writing
- C. avoid using models or discussion
- D. record what they notice in an accessible form and then explain it using evidence

Answer: D

Explanation: Separating evidence from interpretation gives multiple entry points and values reasoning.

Set 19 — Projects, Fieldwork and Science Communication (Q181–Q190)

Projects and fieldwork should be purposeful, safe, evidence-based and followed by communication and reflection.

Q181. A class investigates whether sunlight affects seedling growth. The best design is to —

- A. keep seed type, soil, water and time comparable while changing the light condition and measuring growth
- B. change light, water and soil together
- C. measure only the final tallest plant without records
- D. choose the expected result before planting

Answer: A

Explanation: Controlling relevant conditions makes the light comparison meaningful.

Q182. A learner says every metal object sinks in water. The teacher should—

- A. agree because metals are heavy
- B. test safe examples, compare density and discuss why the observation does not apply universally
- C. ask the learner to copy "metals sink"
- D. avoid testing objects

Answer: B

Explanation: The claim is an overgeneralisation; evidence and density provide a better explanation.

Q183. To investigate shadow length during a day, learners should—

- A. draw one shadow from memory
- B. change the object at every observation
- C. measure the same object's shadow at recorded times and compare the data
- D. observe without recording time

Answer: C

Explanation: Repeated measurements with a constant object link shadow length with time and Sun position.

Q184. A field visit to a pond is scientifically meaningful when learners—

- A. collect organisms without a purpose
- B. treat the visit as only a picnic
- C. write conclusions before seeing the pond
- D. observe selected features, record data, follow safety rules and analyse findings afterwards

Answer: D

Explanation: Purposeful fieldwork joins direct experience with evidence and reflection.

Q185. When studying local weather, a teacher should—

- A. combine learners' observations with instruments, records and discussion of patterns and limits
- B. accept a single person's prediction as fact
- C. avoid measuring temperature or rainfall
- D. teach only global averages

Answer: A

Explanation: Local observations become science when they are systematic, measured and interpreted carefully.

Q186. A strong science project report should include—

- A. only a colourful cover
- B. question, method, observations or data, explanation, limitations and communication of findings
- C. a conclusion with no method
- D. copied information unrelated to the investigation

Answer: B

Explanation: A report makes the inquiry process and evidence transparent.

Q187. An experiment produces an unexpected result. The teacher should—

- A. erase the result because it differs from the prediction
- B. declare the investigation useless immediately
- C. help learners check procedure and records, consider explanations and treat the result as evidence
- D. change the data to match the textbook

Answer: C

Explanation: Unexpected evidence can reveal error, variation or a better question; it should be investigated honestly.

Q188. When observing wildlife for a school project, learners should—

- A. capture animals for convenience
- B. feed every animal to obtain a photograph
- C. share sensitive locations without concern
- D. avoid disturbance, follow ethical guidelines and record observations responsibly

Answer: D

Explanation: Responsible science protects organisms and habitats while collecting evidence.

Q189. Which question best supports science communication after a project?

- A. What evidence supports your conclusion, and how could you communicate it to the community?
- B. Who decorated the title most brightly?
- C. Can you copy the introduction again?
- D. Why must everyone use the same conclusion?

Answer: A

Explanation: Communication should make evidence and implications understandable to an audience.

Q190. A science notebook is most useful when learners—

- A. copy only final definitions
- B. record dated observations, questions, diagrams, data, revisions and reflections
- C. remove every unsuccessful attempt
- D. write answers without describing evidence

Answer: B

Explanation: A notebook documents the development of scientific thinking, not only polished conclusions.

Set 20 — Classroom Scenarios and Integrated Revision (Q191-Q200)

These final scenarios combine science content, inquiry, assessment, inclusion and the most appropriate teacher response.

Q191. A learner claims a water filter works because the filtered water looks clear. The teacher should—

- A. accept clarity as proof of potability
- B. reject every filter without testing
- C. discuss the limit of appearance and design tests for particles or microorganisms with appropriate safety
- D. ask learners to drink the sample

Answer: C

Explanation: Clear appearance does not establish safety; scientific claims require suitable evidence and safe procedures.

Q192. A learner draws a food-chain arrow from a hawk to grass to show where energy begins. The teacher should—

- A. mark it wrong without discussion
- B. say arrows in science have no meaning
- C. ask the learner to memorise the chain only
- D. use organism cards or a diagram to trace food energy from producer to consumer

Answer: D

Explanation: Tracing the feeding relationship clarifies that the arrow represents transfer from food to consumer.

Q193. After heavy rain, a learner proposes planting vegetation to reduce soil loss. The teacher should—

- A. help test the idea with comparable soil trays and observe how roots or cover affect runoff
- B. accept it without evidence
- C. reject it because plants solve no problems
- D. teach only the word erosion

Answer: A

Explanation: A practical comparison connects the proposal with observable evidence about erosion.

Q194. A learner records the Moon's shape as changing each night. The teacher should—

- A. say the Moon actually changes shape
- B. help maintain a dated observation record and use a model to explain changing visible illumination
- C. ignore the observation
- D. give only a diagram without time or position

Answer: B

Explanation: Regular observations and a model distinguish apparent phase from a change in the Moon's physical shape.

Q195. Which question best promotes higher-order thinking about a bulb circuit?

- A. What colour is the wire?
- B. Copy the word circuit
- C. How would the brightness change if one component or connection were changed, and what evidence would support your prediction?
- D. Which learner finished first?

Answer: C

Explanation: The question requires prediction, variable thinking and evidence-based explanation.

Q196. A teacher uses learners' predictions, observations and exit responses to plan the next science lesson. This is—

- A. only summative ranking
- B. a permanent ability label
- C. assessment with no connection to teaching
- D. formative use of assessment

Answer: D

Explanation: The evidence is collected during learning and directly informs the next instruction.

Q197. A learner scores low on a written science test but performs a circuit investigation safely and explains it orally. The teacher should—

- A. use varied evidence and check whether reading or writing barriers obscured the intended understanding
- B. conclude that the learner knows no science
- C. ignore practical performance
- D. lower all science expectations

Answer: A

Explanation: Fair assessment aligns evidence with the construct and recognises suitable demonstrations of understanding.

Q198. A group assigns all measuring and apparatus work to one confident learner. The teacher should—

- A. let the confident learner do all the science
- B. rotate roles, model collaboration and ensure each learner practises the relevant skill
- C. remove group work permanently
- D. grade the group only by speed

Answer: B

Explanation: Role rotation promotes equitable access, skill development and shared responsibility.

Q199. After diagnosing confusion between evaporation and boiling, a suitable remedial step is to—

- A. make learners copy both definitions only
- B. say the terms are interchangeable
- C. compare everyday examples and guided observations of surface evaporation and boiling with evidence
- D. avoid all water investigations

Answer: C

Explanation: Contrasting observations and conditions helps learners build a meaningful distinction.

Q200. Which teacher best reflects CTET Paper-II Science pedagogy?

- A. A teacher who dictates facts and avoids questions
- B. A teacher who values only textbook recall
- C. A teacher who changes results to match the expected answer
- D. A teacher who uses safe inquiry, learner ideas, evidence, varied assessment, inclusion, local context and remedial support.

Answer: D

Explanation: Strong science pedagogy integrates concepts, inquiry, safety, learner diversity, assessment and honest communication of evidence.

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Revision target: *In pedagogy questions, first identify the learner's idea or evidence, then choose the response that preserves the scientific goal, provides an accessible investigation and uses the result to plan the next step.*

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